1 Online Appendix

1.1 Job profiles metadata

The metadata in Table A1 were used to generate role profiles for variant creation; a detailed explanation is provided in the full paper content.

Table A1: Job profiles used for expert role profile generation.

#	File name	Target role	Job title	Source
1	generalist-midlevel.md	Generalist - Mid-Level	Analytical specialist	energyjobsearch.com
2			Downstream sustainability specialist	energyjobsearch.com
3			Energy data portfolio analyst	energyjobsearch.com
4			Energy consultant	euroclimatejobs.com
5			Energy efficiency outreach specialist	rejobs.org
6			Energy resource efficiency sustainability consultant	rejobs.org
7			Market risk manager	euroclimatejobs.com
8			Specialist in demand generation power factors	rejobs.org
9			Sustainability analyst	rejobs.org
10			Sustainability consultant	rejobs.org
11			Technical due diligence expert in biogas	rejobs.org
12	generalist-senior.md	Generalist - Senior	Decarbonization strategy manager	rejobs.org
13			Power & gas market risk analyst	euroclimatejobs.com
14			Program manager	rejobs.org
15			Program manager solar PV	euroclimatejobs.com
16			Project manager solutions generation	energyjobsearch.com
17			Senior market risk manager	euroclimatejobs.com
18			Senior portfolio manager	euroclimatejobs.com
19			Senior product manager	rejobs.org
20			Senior project manager	euroclimatejobs.com
21			Senior project manager	rejobs.org
22			Senior renewable energy consultant	euroclimatejobs.com
23	normative-midlevel.md	Normative - Mid-Level	Associate general counsel	rejobs.org
24			Enterprise architecture practice & governance lead	energyjobsearch.com
25			Internal communication & engagement officer	rejobs.org

Table A1 – Continued

#	File-Name	Target-Role	Job-Title	Source
26			Licensing & permitting officer	energyjobsearch.com
27			Manager of community energy planning & policy	rejobs.org
28			Operational superviser	energyjobsearch.com
29			Project developer	rejobs.org
30			Strategic sourcing manager	energyjobsearch.com
31			Sustainability air advisor	energyjobsearch.com
32	normative-senior.md	Normative - Senior	Detail-oriented excellence manager	energyjobsearch.com
33			Director technical services	rejobs.org
34			Environment regulatory & socioeconomic advisor	energyjobsearch.com
35			Industrial decarbonization	rejobs.org
36			Manager BESS energy product	euroclimatejobs.com
37			Project development manager	euroclimatejobs.com
38			Project development director	rejobs.org
39			Senior expert in energy solutions	euroclimatejobs.com
40			Senior permitting & regulatory officer	rejobs.org
41	sme-midlevel.md	SME - Mid-Level	Automation engineer	rejobs.org
42			Automation Engineer	energyjobsearch.com
43			Automation Engineer	energyjobsearch.com
44			BESS project engineer	euroclimatejobs.com
45			electrical Engineer	euroengineerjobs.com
46			Electrical Engineer	euroengineerjobs.com
47			Electrical Engineer	euroengineerjobs.com
48			Electrical Engineer	energyjobsearch.com
49			Field Service Engineer (EV Chargers)	energyjobsearch.com
50			Low Voltage Systems Engineering Manager	energyjobsearch.com
51			Multi Skilled Engineer (Electrical Bias)	energyjobsearch.com
52			Solar Lead Engineer	rejobs.org
53	sme-senior.md	SME - Senior	BESS manager-design, engineering & cost estimation	rejobs.org
54			BESS project engineer	euroclimatejobs.com
55			Commissioning engineer- HVDC protection	rejobs.org
56			Electrical engineer	energyjobsearch.com
57			HV technician	energyjobsearch.com

Table A1 – Continued

#	File-Name	Target-Role	Job-Title	Source
58			Lead electronics & Electrical Engineer	euroengineerjobs.com
59			Multi skilled engineer	energyjobsearch.com
60			Onshore delivery engineering	energyjobsearch.com
61			Project manager BESS	euroengineerjobs.com
62			Senior commissioning and maintenance electrical	energyjobsearch.com
63			Senior High Voltage technician - DC protection	rejobs.org

Note: Each *file name* represents the files that were used for the generation of role profiles. Some job titles across roles seem to be repeated, but they vary in content because they were posted by different sources (i.e., companies) and, in some cases, accessed through different websites.

Acronyms (Table A1.): DC (Direct Current); BESS (Battery Energy Storage System); HV (High-Voltage); EV (Electric Vehicle); PV (Photo-Voltaics).

1.2 Cosine similarity on role-profiles

The initially generated profiles (at multiple runs) were first assessed for similarity prior to selection, as shown in Figure 1, and later assessed for similarity between categories' role-profiles as shown in Figure 2

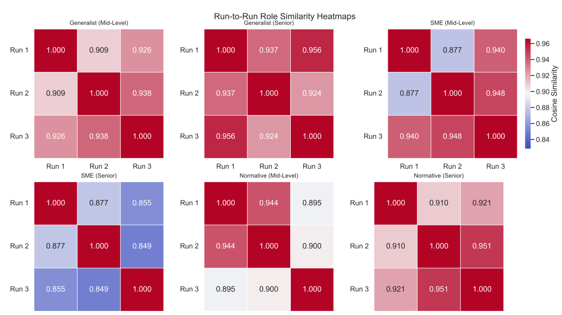

Fig. 1. Heatmap on similarity scores per variant between the three runs

1.3 Prompt composition to LLM variants

This section describes prompt composition as shown in Figure 3. Whereby, the structure comprises—project descriptions, descriptions of the target demonstration sites, the elicitation process and instructions, and the question. All of these parts were compiled together in a Python script and provided to the variants during the generation step.

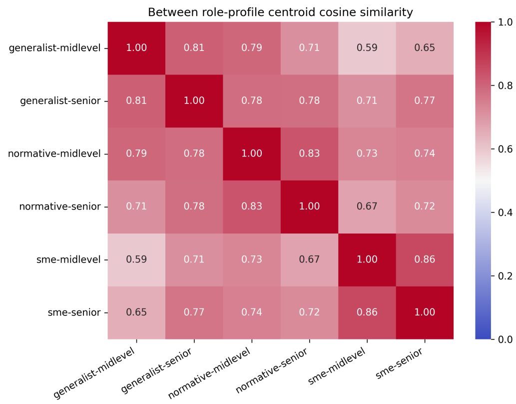

Fig. 2. Heatmap on similarity scores within-category role-profiles.

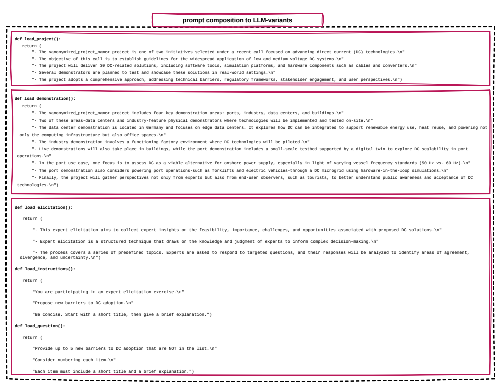

Fig. 3. Prompt composition to LLM-variants.

2 Cluster details

This section contains details on the clusters generated by the 24 LLM variants from the barriers. The details are as shown in Table A2, sorted from the cluster with the biggest size to the lowest.

Table A2. Cluster-level summary of HDBSCAN results with GPT-generated and human-assigned labels.

ID	Size	GPTClusterName	HumanLabels
1	197	cybersecurity vulnerabilities in dc microgrids	Cybersecurity vulnerabilities in DC microgrids
10	109	dc supply chain constraints	DC supply chain constraints
17	79	public awareness and acceptance	Public safety, awareness and acceptance issues
25	61	dc ac interoperability challenges	DC interoperability and standardization challenges
26	50	dc standardization and interoperability	DC interoperability and standardization challenges
21	47	regulatory uncertainty and compliance	Regulatory inconsistency and compliance barriers
27	47	dc interoperability standards	DC interoperability and standardization challenges
16	36	public safety perception concerns	Public safety, awareness and acceptance issues
24	36	existing ac grid integration	Interoperability challenges with existing AC infrastructure
12	35	retrofit costs and complexity	Retrofit costs and compatibility complexity
20	33	legacy infrastructure compatibility	Retrofit costs and compatibility complexity
30	31	regulatory and standards barriers	Regulatory inconsistency and compliance barriers
28	30	dc interoperability standards	DC interoperability and standardization challenges
0	28	dc safety standards gap	Challenges in DC safety standardization
29	27	regulatory inconsistency and compliance barriers	Regulatory inconsistency and compliance barriers
8	25	economic uncertainty and roi	Economic uncertainty and ROI
4	24	dc interoperability standardization	DC interoperability and standardization challenges
3	23	regulatory uncertainty and hurdles	Regulatory inconsistency and compliance barriers
22	23	retrofit costs and complexity	Retrofit costs and compatibility complexity
2	22	skilled workforce shortage	Shortage in skilled workforce
6	19	lack of dc standardization	DC interoperability and standardization challenges
13	17	interoperability with existing ac infrastructure	Interoperability challenges with existing AC infrastructure
23	17	retrofit costs and disruption	Retrofit costs and compatibility complexity
14	15	public awareness and acceptance	Public safety, awareness and acceptance issues
19	15	limited dc hardware availability	DC component availability and durability
18	14	lack of standardized dc components	DC interoperability and standardization challenges
15	12	public safety perception concerns	Public safety, awareness and acceptance issues
7	11	ac grid integration complexity	Interoperability challenges with existing AC infrastructure
9	11	high upfront transition costs	High upfront transition cost
5	10	dc standardization gaps	DC interoperability and standardization challenges
11	10	dc component cost availability	DC component availability and durability

Acronyms: ROI —> Return on Investment; DC —> Direct Current; AC —> Alternating Current.

Shared sub-themes raised from GPT-naming stage: DC interoperability standards (ID: 27, 28); public awareness and acceptance (ID: 14, 17); public safety perception concerns (ID: 15, 16); retrofit costs and complexity (ID: 12, 22).